

Software Requirements Specification (SRS)



Project Name: **TrackDel: Supply Chain Delivery Performance Dashboard**

Version: 1.0

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1. INTRODUCTION

TrackDel is a data analytics project developed to analyze supply chain delivery performance using Python, SQL, and Power BI. The system helps identify delivery delays, monitor order status, evaluate shipping performance, and measure logistics efficiency.

The dashboard provides business insights that help understand the factors responsible for delayed deliveries and improve overall supply chain operations.

1.1 Purpose

The purpose of TrackDel is to:

- Monitor delivery performance.
- Identify delayed orders.
- Analyze causes of delivery delays.
- Evaluate shipping performance.
- Track order status.
- Improve logistics efficiency through data-driven insights.

1.2 Scope

Features:

- Analyze supply chain data.
- Calculate delivery-related KPIs.
- Track delayed and on-time deliveries.
- Analyze shipping modes and order status.
- Provide interactive Power BI dashboards.

Users:

- Supply Chain Manager
- Logistics Manager
- Business Analyst

Limitations:

- Uses historical dataset only.
- No real-time tracking.
- No online order management functionality.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
KPI	Key Performance Indicator
SQL	Structured Query Language
EDA	Exploratory Data Analysis
BI	Business Intelligence
Delay Days	Difference between actual and scheduled shipping days
On-Time Delivery	Order delivered within scheduled time

2. OVERALL DESCRIPTION

TrackDel processes supply chain data and generates analytical dashboards for performance monitoring.

The system focuses on delivery delays, shipping performance, order status, and logistics efficiency.

2.1 Product Perspective

The system consists of:

- Supply Chain Dataset
- SQL Database
- Data Cleaning (Python)
- KPI Calculation
- Power BI Dashboard
- Business Insights

2.2 Product Functions

The system shall:

- Import supply chain dataset.
- Store data in SQL database.
- Clean and preprocess data using Python.
- Calculate delivery KPIs.
- Identify delayed orders.
- Analyze shipping performance.
- Analyze order status.
- Analyze supplier performance.
- Generate Power BI dashboards.

2.3 User Characteristics

End Users

- Supply Chain Manager – Monitors delivery performance and supplier performance.
- Logistics Manager – Analyzes shipping performance and logistics efficiency.
- Business Analyst – Reviews dashboards, analyzes trends, and generates reports.

Administrator

- Project Developers – Responsible for data preparation, maintenance, and dashboard updates.

2.4 Constraints

Technical Constraints

- Dataset quality affects results.
- Requires Power BI Desktop.
- Requires Python environment.

Legal Constraints

- Dataset must be used according to its license and usage terms.
- No unauthorized distribution or misuse of the dataset.

Business Constraints

- Analysis depends on available dataset columns.
- Historical analysis only.

3. FUNCTIONAL REQUIREMENTS

ID	Requirement
FR-01	System shall import supply chain dataset.
FR-02	System shall clean missing and duplicate records.
FR-03	System shall calculate delivery KPIs.
FR-04	System shall identify delayed orders.
FR-05	System shall analyze shipping modes.
FR-06	System shall analyze order status.
FR-07	System shall generate interactive dashboards.
FR-08	System shall provide filtering by region and shipping mode.
FR-09	System shall analyze supplier performance.

4. NON-FUNCTIONAL REQUIREMENTS

Performance: Dashboard should load within a few seconds.

Security: Dataset access should be limited to authorized users.

Scalability: System should support larger datasets.

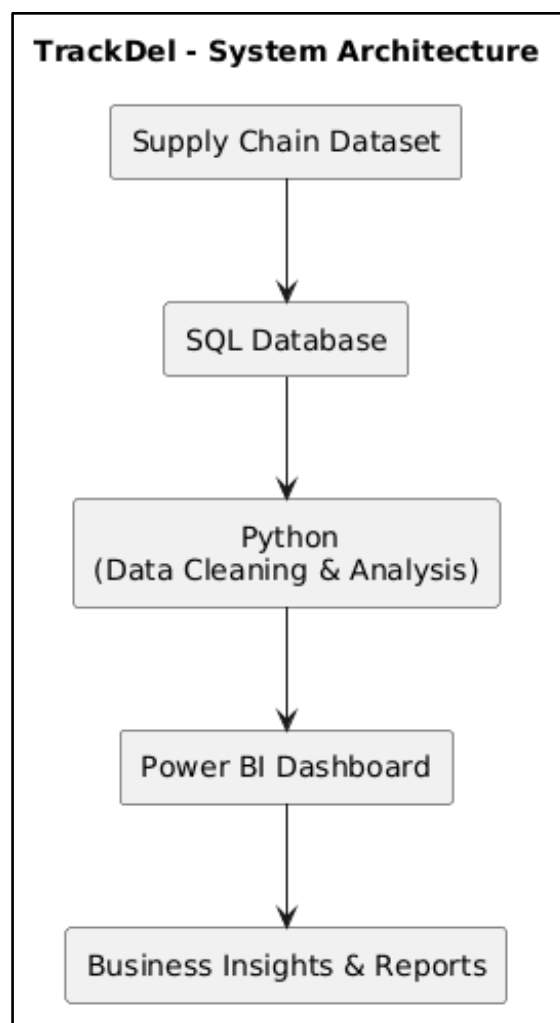
Usability: Dashboard should be easy to understand.

Reliability: Results should be consistent for the same dataset.

5. SYSTEM ARCHITECTURE

Architecture Flow:

Supply Chain Dataset -> SQL Database -> Python (Data Cleaning & Analysis) ->
Power BI Dashboard -> Business Insights & Reports



Modules:

1. Data Storage Layer (SQL)
2. Data Processing Layer (Python)
3. Visualization Layer (Power BI)

6. DATABASE DESIGN

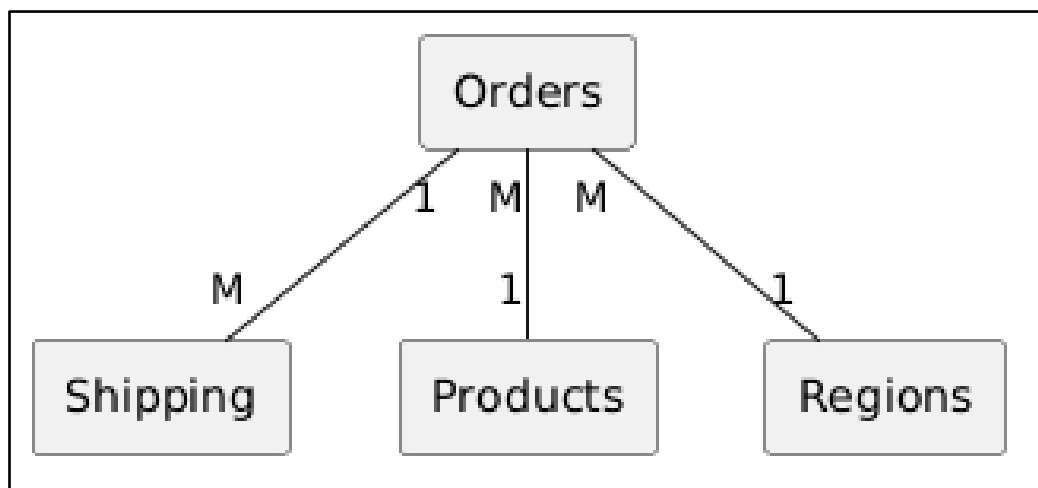
The TrackDel system uses a relational database to store and analyze supply chain data. The database is divided into four main tables: Orders, Shipping, Products, and Regions. These tables help organize data efficiently and support delivery performance analysis.

Tables:

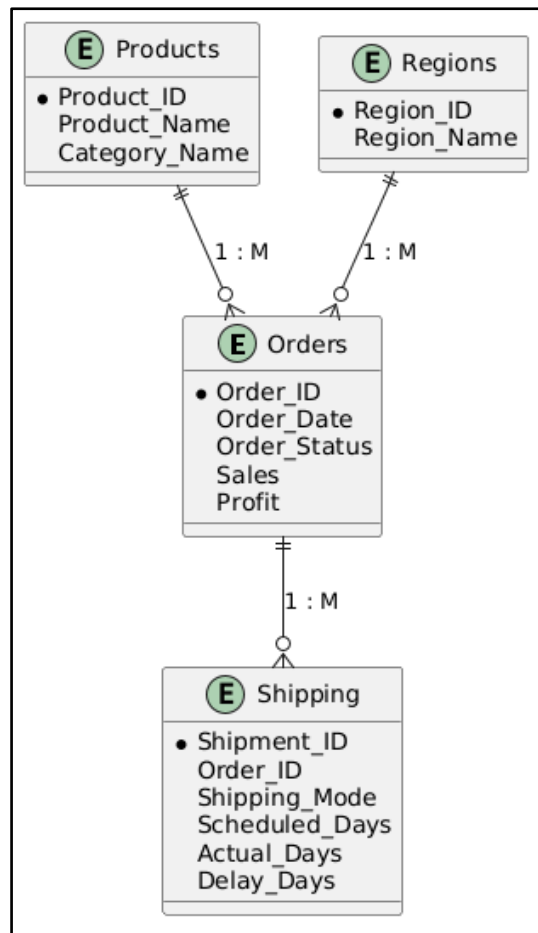
- **Orders:** Order_ID, Order_Date, Order_Status, Sales, Profit
- **Shipping:** Shipment_ID, Order_ID, Shipping_Mode, Scheduled_Days, Actual_Days, Delay_Days
- **Products:** Product_ID, Product_Name, Category_Name
- **Regions:** Region_ID, Region_Name

Relationships:

- One order can have shipping information associated with it.
- Orders are linked to products for category analysis.
- Orders are linked to regions for geographical performance analysis.



ER Diagram:



7. USE CASES

The TrackDel system allows users to monitor and analyze supply chain performance through interactive dashboard modules. Users can analyze delivery delays, evaluate supplier performance, analyze shipping performance, monitor order status, assess logistics efficiency, and generate analytical reports to support decision-making.

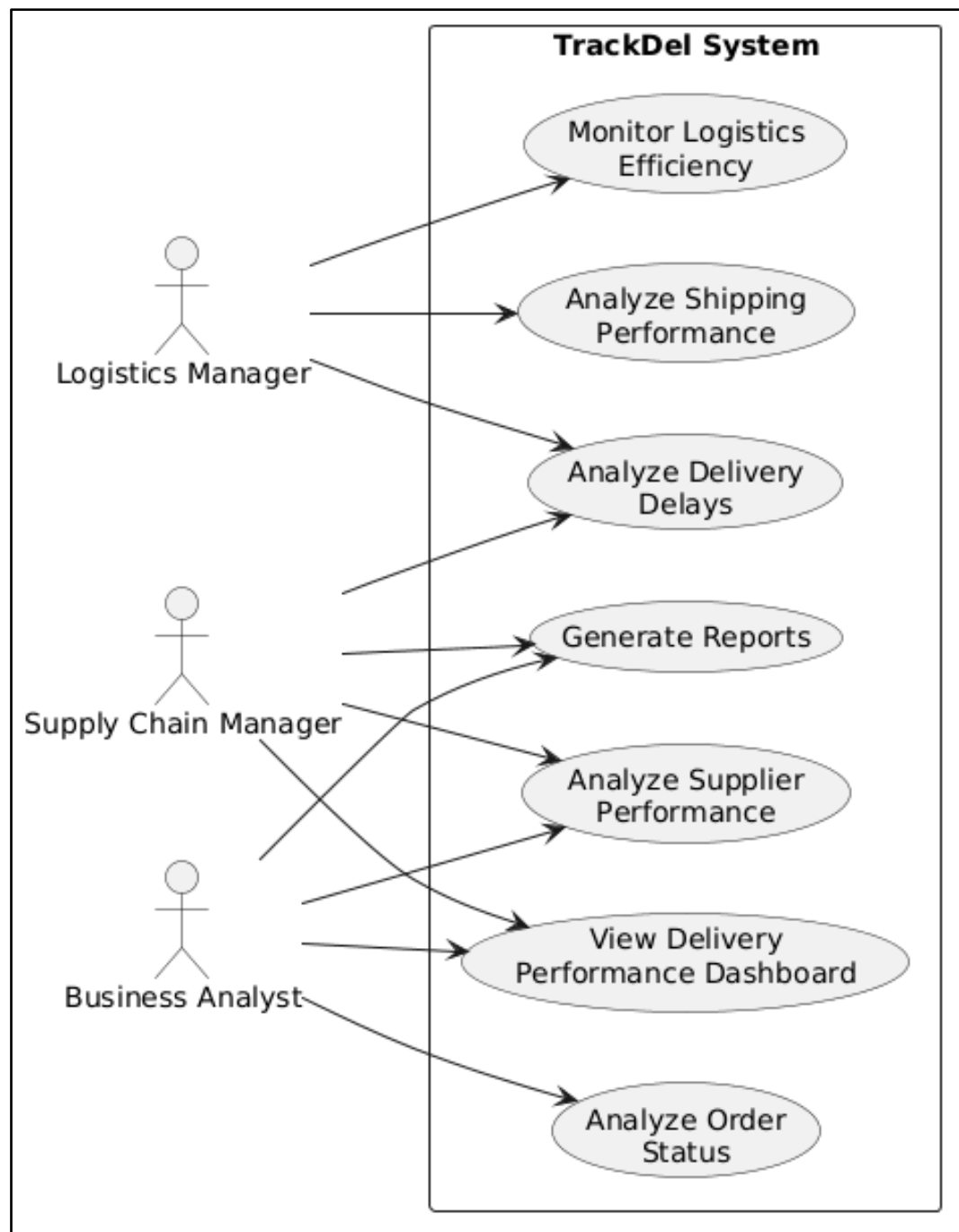
Actors:

- Supply Chain Manager
- Logistics Manager
- Business Analyst

Main Use Cases:

- View Delivery Performance Dashboard
- Analyze Delivery Delays
- Analyze Supplier Performance
- Analyze Shipping Performance
- Analyze Order Status
- Monitor Logistics Efficiency
- Generate Reports

Use Case Diagram:



8. EXTERNAL INTERFACE REQUIREMENTS

User Interface

The system provides an interactive Power BI dashboard containing:

- KPI Cards
- Bar Charts
- Pie/Donut Charts
- Line Charts
- Slicers and Filters

Software Interface

The system integrates with the following software components:

- SQL Database
- Python (Data Cleaning and Analysis)
- Power BI (Data Visualization and Reporting)

Hardware Interface

The system requires:

- Computer or Laptop
- Storage for dataset and project files
- Display device for dashboard visualization

9. SECURITY REQUIREMENTS

Authentication: Access to project files and dashboards should be limited to authorized users.

Authorization: Only project developers are allowed to modify the dataset, SQL database, and dashboard components.

Encryption: No encryption mechanism is implemented in the current version of the TrackDel system.

10. TESTING REQUIREMENTS

Unit Testing: Verify KPI calculations.

Integration Testing: Verify SQL, Python, and Power BI integration.

System Testing: Verify complete dashboard functionality.

11. FUTURE ENHANCEMENTS

- Real-time shipment tracking.
- Predictive delay analysis using Machine Learning.
- Web-based dashboard deployment.

12. REFERENCES

- Python Documentation
- SQL Documentation
- Microsoft Power BI Documentation
- Supply Chain Dataset (CSV File)